HAROL MAXILAN

DATA ENGINEERING INTERN | DATA ANALYTICS

+94 70 521 8792 maxilanharrol@gmail.com harol-maxilan Max-i26 Max-i26

Vavuniya, Sri Lanka

Professional Summary

Data Science undergraduate at Sabaragamuwa University of Sri Lanka with hands-on project experience in data processing, pipeline development, data quality validation, and analytical workflows using Python, SQL, and Pandas. Experienced in developing a medallion-architecture data pipeline, automating data validation checks, processing large datasets, and building data-driven applications. Familiar with Microsoft Excel and Power BI, with a strong interest in ETL processes, database management, and developing reliable analytical data pipelines. Eager to contribute to data engineering operations while strengthening practical industry skills.

Education

B.Sc. (Hons) in Data Science | 2024 – 2028 (Expected), *Sabaragamuwa University of Sri Lanka — Faculty of Computing*
B.Sc. (Hons) in Data Science | 2024–2028 (Expected)
Sabaragamuwa University of Sri Lanka – Faculty of Computing

Relevant Coursework: Database Systems, Data Warehousing, Data Science, Statistics, Computer Programming, Machine Learning, Artificial Intelligence

Awards

Nexora Ideathon : 3rd Place
Sabaragamuwa University of Sri Lanka

Cipher Hackathon : Finalist
IIT

J'PuraXtreme 3.0 Hackathon : Finalist
University of Sri Jayewardenepura

CodeRally 7.0 Hackathon : Finalist
IIT

Code Arena : Finalist
KDU

Projects

Syndicate_DataStrom — Lakehouse Pipeline & Demand Prediction Model, *Technologies: Python, Pandas, Scikit-learn, OpenStreetMap API, Medallion Architecture*

- Designed and implemented a Bronze–Silver–Gold data pipeline using Medallion Architecture to process data for demand prediction across 20,000 retail outlets.
- Implemented automated data quality checks for duplicate records, referential integrity, and format validation, with invalid records quarantined in the Silver layer.
- Developed automated data extraction workflows to retrieve geospatial Point of Interest (POI) data through the OpenStreetMap Overpass API and generate spatial features in the Gold layer.

F1 Super Intelligence System — Data Analytics & Precomputation Pipeline, *Technologies: Python, Pandas, Plotly, Pytest, Batch Processing, Data Auditing*

- Developed a batch-processing pipeline to aggregate and cache Formula 1 telemetry datasets, supporting sub-second rendering for interactive analytical dashboards.
- Implemented dataset completeness auditing across 24 rounds and developed a 56-test suite covering data anomalies, race strategies, and telemetry integrity.
- Built statistical processing workflows to analyse driver pace and consistency, transforming complex data into interactive visualizations.

Friday — Automated Machine Learning Pipeline, *Technologies: Python, Pandas, Scikit-learn, FastAPI, Google Generative AI*

- Developed an automated machine learning pipeline with a FastAPI backend to support data-driven predictive modelling.
- Built data-processing workflows using Pandas and Scikit-learn to transform raw datasets into statistical analyses and meaningful insights.
- Integrated Google Generative AI to generate analytical reports from processed data.

TECHNICAL SKILLS

Programming Languages
Python, SQL, Java, JavaScript, PHP

Data Engineering & Processing
Data preprocessing, data transformation, ETL concepts, batch processing, data validation, data quality checks, duplicate detection, referential integrity, data pipeline development

Data Analysis & Visualization
Pandas, NumPy, Matplotlib, Plotly, exploratory data analysis (EDA), statistical analysis, feature engineering

Databases
MySQL, MariaDB

Development & Tools
Git, GitHub, Jupyter Notebook, VS Code, Streamlit

Machine Learning
Classification, Regression, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation

References

Ms.T.Kayalvizhy, *Lecturer,*
Sabaragamuwa University of Sri Lanka
kayalvizhy@foc.sab.ac.lk, 0754376923

Mr. Mahesh Yogarajan, *CEO,*
Lanka.Tax, 247D, R. A. De Mel Mawatha, Colombo 3.
mahesh@yogarajan.com, +94-777-739-739

Leadership & Volunteer Experience

AWS Logistics and Operations Lead
10/2026 – Present
Sabaragamuwa University of Sri Lanka | Current

Club Cricket Team Captian
10/2025
VSC, Vavuniya

Clash of Pawns Chess Competition Mentor
06/2025
AUSDAV, Vavuniya

Zone C University Ambassador – IEEE INSL
04/2025
Sabaragamuwa University of Sri Lanka

Pentathalon 2.0 Logistics Lead
12/2023
AUSDAV, Vavuniya